

Sub-2Sub-2Sub-3

CAGR  
**+26.5%**  
annualised

Max Drawdown  
**-13.0%**  
worst peak-to-trough

Sharpe  
**1.67**  
annualised

MAR  
**2.04**  
CAGR / MaxDD

Total Return  
**+700%**  
8.8 years

Sortino  
**2.71**  
downside-adjusted

Volatility  
**14.7%**  
annualised

Longest DD  
**221d**  
median spell 6d

Equity curve — portfolio vs sleeves (log scale, 1x)



Portfolio drawdown



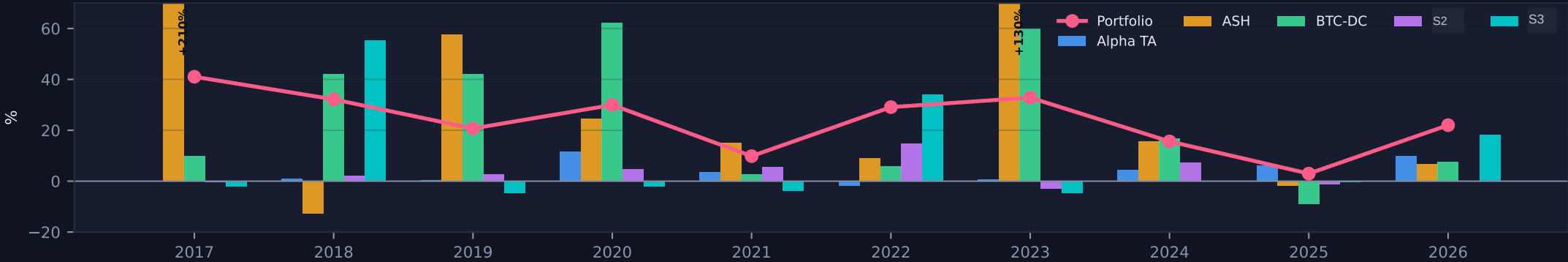
| Sleeve               | Notional wt | Risk share | CAGR   | MaxDD  | Sharpe | MAR  | Longest DD |
|----------------------|-------------|------------|--------|--------|--------|------|------------|
| Alpha TA             | 86%         | 14%        | +4.0%  | -6.4%  | 0.76   | 0.62 | 773d       |
| ASH                  | 29%         | 54%        | +39.1% | -38.5% | 1.11   | 1.02 | 560d       |
| BTC-DC               | 14%         | 13%        | +24.9% | -17.5% | 1.13   | 1.42 | 439d       |
| Sub-2                | 57%         | 8%         | +3.6%  | -11.0% | 0.57   | 0.33 | 550d       |
| Sub-3                | 50%         | 10%        | +8.5%  | -12.1% | 1.05   | 0.70 | 1228d      |
| Sub-4 (retired)      | —           | —          | +4.2%  | -40.1% | 0.31   | 0.10 | 1943d      |
| BTC buy & hold       | —           | —          | +35.9% | -82.0% | 0.80   | 0.44 | 1095d      |
| PORTFOLIO (Option B) | 186%        | 100%       | +26.5% | -13.0% | 1.67   | 2.04 | 221d       |

Notional weight is exposure; risk share is the contribution to portfolio volatility. A low-drawdown sleeve can carry a large notional weight cheaply — ASH is the largest RISK position despite the smallest weight.

# Yearly & Monthly Breakdown

per-year returns by sleeve, portfolio monthly grid, and return distribution

Annual return by sleeve (1x, unweighted) — bars clipped at +70%



Portfolio monthly returns (%)

|      |       |      |      |      |       |       |      |      |      |       |       |       |       |
|------|-------|------|------|------|-------|-------|------|------|------|-------|-------|-------|-------|
| 2017 | .     | .    | .    | .    | .     | .     | .    | .    | +0.6 | +9.9  | +9.8  | +16.2 | +41.0 |
| 2018 | +7.6  | -0.1 | +6.3 | +3.7 | +3.9  | +2.7  | +3.4 | +6.6 | -6.9 | -2.3  | +11.8 | -6.6  | +32.2 |
| 2019 | +0.8  | +1.8 | +1.0 | +3.9 | +13.5 | +3.1  | +0.1 | -1.8 | -1.4 | -0.5  | -0.2  | -0.6  | +20.7 |
| 2020 | +5.6  | +0.1 | +1.1 | +1.4 | +1.1  | -5.0  | +5.4 | -1.4 | +0.1 | +14.5 | +3.0  | +1.7  | +30.0 |
| 2021 | +2.3  | +1.1 | -0.7 | -0.4 | -3.0  | +0.8  | -0.4 | +5.6 | -4.3 | +6.0  | +1.2  | +1.6  | +9.8  |
| 2022 | +5.5  | +3.0 | +3.5 | +0.4 | +8.9  | +9.8  | -2.3 | -3.2 | -1.2 | -3.1  | +3.8  | +1.6  | +29.1 |
| 2023 | +15.1 | -0.7 | +2.9 | +0.7 | -1.0  | +5.4  | -1.0 | -3.5 | +0.3 | +8.7  | -0.4  | +3.6  | +32.8 |
| 2024 | -0.4  | +5.1 | -0.0 | -0.1 | +0.1  | -1.5  | +2.3 | +2.0 | +0.7 | -1.1  | +6.1  | +1.6  | +15.6 |
| 2025 | -4.8  | +5.9 | -0.3 | -0.2 | +7.2  | +0.3  | -1.6 | -1.7 | -1.8 | -0.2  | +2.0  | -1.2  | +3.0  |
| 2026 | +8.5  | +4.1 | -1.5 | -2.9 | +1.9  | +11.6 | -0.1 | -0.5 | .    | .     | .     | .     | +22.0 |
|      | Jan   | Feb  | Mar  | Apr  | May   | Jun   | Jul  | Aug  | Sep  | Oct   | Nov   | Dec   | Year  |

Daily return distribution (active days)



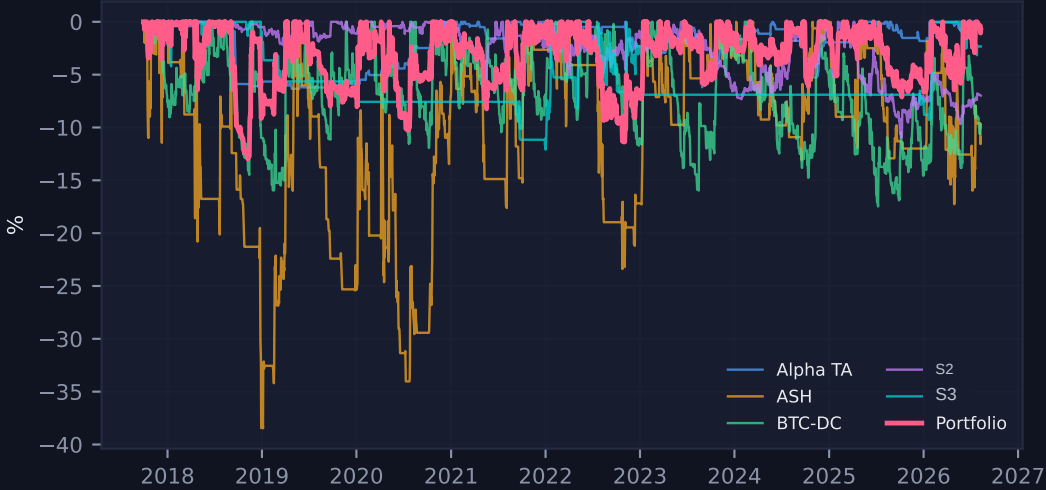
# Trade Statistics, Risk & Correlation

per-trade behaviour of the signal sleeves, drawdown profile, and sleeve correlations

## Per-trade statistics (full history, 1x, net of costs)

| Sleeve | Trades | /yr | Win rate | Profit factor | Avg trade | Avg win | Avg loss | Best   | Worst  | Avg hold |
|--------|--------|-----|----------|---------------|-----------|---------|----------|--------|--------|----------|
| ASH    | 379    | 82  | 41.7%    | 2.08          | +1.06%    | +4.91%  | -1.68%   | +32.1% | -10.8% | 2.1d     |
| BTC-DC | 373    | 81  | 37.0%    | 1.55          | +1.13%    | +8.57%  | -3.24%   | +58.8% | -20.3% | 4.6d     |

## Drawdown — portfolio vs sleeves



## Sleeve correlation (daily, full period)

|     |      |       |      |       |       |
|-----|------|-------|------|-------|-------|
| ATA | 1.00 | 0.12  | 0.14 | 0.02  | 0.03  |
| ASH | 0.12 | 1.00  | 0.42 | -0.05 | -0.05 |
| DC  | 0.14 | 0.42  | 1.00 | 0.17  | 0.26  |
| S2  | 0.02 | -0.05 | 0.17 | 1.00  | 0.26  |
| S3  | 0.03 | -0.05 | 0.26 | 0.26  | 1.00  |
|     | ATA  | ASH   | DC   | S2    | S3    |

## Risk summary — portfolio

| Risk metric               | Value  | Risk metric           | Value                |
|---------------------------|--------|-----------------------|----------------------|
| Annualised volatility     | 14.7%  | Longest drawdown      | 221 days             |
| Best day                  | +8.05% | Median drawdown spell | 6 days               |
| Worst day                 | -5.24% | Sortino ratio         | 2.71                 |
| Positive days (of active) | 48.4%  | Design DD budget      | -8.5% (ceiling -10%) |

## Results by window

| Window                              | CAGR   | MaxDD  | Sharpe | Total  |
|-------------------------------------|--------|--------|--------|--------|
| Full tested period 2017-2026 (8.8y) | +26.5% | -13.0% | 1.67   | +700%  |
| Out-of-sample only 2022-2026 (4.6y) | +21.9% | -11.4% | 1.56   | +149%  |
| BTC buy & hold 2017-2026            | +35.9% | -82.0% | 0.80   | +1416% |

The headline figures on page 1 use the full tested period. The 2022-2026 row is the stricter out-of-sample cut; ASH and BTC-DC parameters were chosen knowing the earlier history, so pre-2022 is not strictly out-of-sample. Both are shown so the reader can judge; neither was selected for how it looks.